

Comparative Sentiment Analysis of Political and Social Discourse Across News Media and Online Communities

Programming for data analysis

Ammi Susana Pineda Guzman - 20221020064
David Santiago Buitrago Cerquera - 20221020085
Cristhian Samuel Truque Cabrejo - 20221020071

Systems Engineering

University Distrital Francisco José de Caldas

Bogotá D.C, Colombia

GitHub Project Link: <https://github.com/SantyaGo710/dataAnalysis>

I. EXECUTIVE SUMMARY

This workshop extends the data engineering pipeline developed in previous stages by incorporating a Gold Layer focused on governance metrics, storytelling analytics, and business-oriented insights. The project processes data collected from Reddit and La Silla Vacía through a multi-layer architecture composed of Bronze, Silver, and Gold stages.

The Bronze layer stores the raw data collected from both sources, while the Silver layer applies data cleaning, normalization, deduplication, and quality validation processes. Building upon these refined datasets, the Gold layer generates aggregated datasets designed to support analytical reporting, governance monitoring, and storytelling-driven exploration.

To achieve this objective, Apache Airflow orchestrates the entire workflow, while PySpark performs scalable transformations and aggregations. The final outputs include governance indicators, content volume trends, keyword frequency analysis, author activity metrics, and engagement trends based on Reddit scores.

The resulting datasets provide a structured foundation for data governance assessment and exploratory analysis, demonstrating how data engineering pipelines can transform raw web data into valuable analytical assets.

II. EVOLUTION FROM WORKSHOP 2 TO WORKSHOP 3

One of the main objectives of Workshop 3 was to improve the existing data pipeline based on observations obtained during previous experimentation phases. The transition from Workshop 2 to Workshop 3 focused on increasing data quality, improving compatibility with PySpark processing, and strengthening validation mechanisms before data reached the Gold layer.

Several enhancements were incorporated into the Silver processing stage. First, duplicate detection was strengthened by using a more restrictive comparison strategy for Reddit data. Second, additional validation rules were introduced to

eliminate records with invalid URLs or insufficient textual content. These controls reduced the presence of low-quality records and improved the reliability of downstream analytics.

Furthermore, timestamp handling was standardized using microsecond precision formats compatible with PySpark and Parquet processing. Timezone normalization was also incorporated to guarantee consistency across heterogeneous data sources.

These improvements contributed directly to the quality of the Gold datasets and enabled the implementation of advanced governance and storytelling analyses.

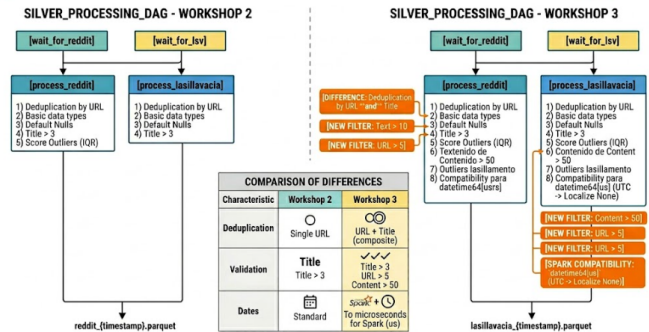


Fig. 1. Silver DAG of workshop 2 VS Silver DAG of workshop 3

III. PIPELINE IMPROVEMENTS BASED ON EXPERIMENTATION

During the development of the pipeline, several experimental iterations were performed to improve data quality, processing consistency, and analytical reliability. The transition from Workshop 2 to Workshop 3 introduced additional validation and normalization mechanisms based on issues identified during exploratory analysis and intermediate processing stages.

The primary objective of these improvements was to ensure that only high-quality and analytically useful records reached

the Gold layer. This was especially important because the final governance metrics and storytelling aggregations depended heavily on the integrity and consistency of the Silver datasets.

A. Enhanced Duplicate Detection

In Workshop 2, duplicate removal for Reddit data was based exclusively on URL comparison. Although this approach eliminated most repeated records, some inconsistencies were still observed when similar posts shared duplicated content with slight variations in titles or metadata.

To address this limitation, Workshop 3 implemented a stricter deduplication strategy by combining URL and title validation.

```
Workshop 2:
df.drop_duplicates(subset=["url"])
Workshop 3:
df.drop_duplicates(subset=["url", "title"])
```

This enhancement improved record uniqueness and reduced the probability of duplicated analytical contributions in downstream aggregations.

B. URL Validation Improvements

During experimentation, some records were identified with incomplete or invalid URLs. Since URLs are fundamental for traceability and governance auditing, additional validation rules were introduced.

```
df = df[df["url"].str.len() > 5]
```

This filtering step ensured that all retained records contained valid references to their original sources.

C. Minimum Content Validation

Exploratory analysis revealed the existence of records containing extremely short textual content that contributed little analytical value. As a result, minimum content thresholds were introduced to guarantee richer and more meaningful textual datasets.

```
Reddit Validation:
df = df[df["text"].str.len() > 10]
La Silla Vacía Validation:
df = df[df["contenido"].str.len() > 50]
```

These rules significantly improved the quality of the corpus used for keyword extraction, storytelling analysis, and descriptive statistics.

D. Timestamp Standardization and PySpark Compatibility

One of the most important improvements introduced in Workshop 3 was the standardization of timestamp precision for compatibility with PySpark and Parquet processing.

```
.astype("datetime64[us]")
```

Using microsecond precision ensured reliable interoperability between Pandas, Parquet serialization, and PySpark transformations executed in the Gold layer.

Additionally, timezone normalization was incorporated to avoid inconsistencies between heterogeneous data sources.

```
utc=True
dt.tz_localize(None)
```

These improvements guaranteed consistent temporal analysis across all aggregated datasets.

E. Data Cleaning and NLP Normalization

The Silver layer also maintained lightweight NLP preprocessing techniques to normalize textual content before analytical processing. The implemented cleaning process included:

lowercase normalization,
URL removal,
punctuation cleaning,
whitespace normalization,
stopword removal,
token filtering.

This preprocessing strategy simplified downstream keyword extraction and reduced textual noise during storytelling aggregation.

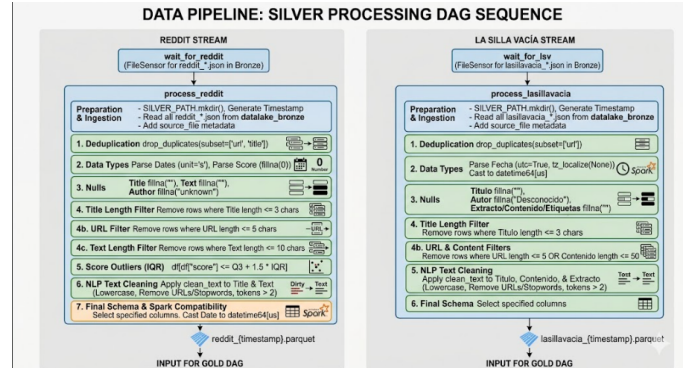


Fig. 2. Silver processing DAG sequence

IV. DATA QUALITY ASSESSMENT AND FINDINGS

A comprehensive data quality assessment was performed to evaluate the reliability, completeness, and analytical usability of the processed datasets. Multiple statistical analyses and validation procedures were applied using the data_quality.ipynb notebook.

A. Dataset Overview

After Silver-layer processing, the final datasets contained the following number of records:

These datasets provided sufficient volume and diversity for governance analysis and storytelling-oriented aggregations.

Source	Records	Attributes
Reddit	243	9
La Silla Vacía	75	12

TABLE I

SUMMARY OF RECORDS AND ATTRIBUTES BY SOURCE.

B. Null Value Analysis

One of the most relevant findings of the quality assessment was the absence of missing values in all critical attributes after Silver-layer processing.

```
print("=== Null rates Reddit ===")
null_rates = (df_reddit.isnull().sum() / len(df_reddit) * 100).round(2)
print(null_rates)

=== Null rates Reddit ===
title      0.0
title_clean 0.0
text       0.0
text_clean 0.0
author     0.0
date       0.0
score      0.0
url        0.0
source_file 0.0
dtype: float64
```

Fig. 3. Null rates La Silla Vacía

```
print("=== Null rates La Silla Vacía ===")
null_rates_lsv = (df_lsv.isnull().sum() / len(df_lsv) * 100).round(2)
print(null_rates_lsv)

=== Null rates La Silla Vacía ===
titulo      0.0
titulo_clean 0.0
autor       0.0
fecha       0.0
extracto    0.0
extracto_clean 0.0
contenido   0.0
contenido_clean 0.0
etiquetas   0.0
url         0.0
fuente      0.0
source_file 0.0
dtype: float64
```

Fig. 4. Null rates Reddit

These results demonstrate that the implemented validation and cleaning strategies successfully eliminated incomplete records before analytical processing.

C. Duplicate Analysis

Duplicate detection was evaluated after the enhanced deduplication strategies introduced in Workshop 3.

```
print("=== Duplicate analysis ===")
dup_reddit = df_reddit.duplicated(subset=["url"]).sum()
dup_lsv = df_lsv.duplicated(subset=["url"]).sum()
print(f"Reddit duplicates: {dup_reddit} ({dup_r (variable) dup_lsv: Any .2f}%)"
print(f"La Silla Vacía duplicates: {dup_lsv} ({dup_lsv/len(df_lsv)*100:.2f}%)"

[8]

=== Duplicate analysis ===
Reddit duplicates: 0 (0.00%)
La Silla Vacía duplicates: 0 (0.00%)
```

Fig. 5. Duplicate analysis

The absence of duplicates confirms the effectiveness of the implemented filtering mechanisms and guarantees that aggregated metrics are not artificially inflated by repeated records.

D. Text Length Analysis

Textual content analysis revealed significant variability in article sizes, especially for La Silla Vacía content.

```
print("=== Text length statistics La Silla Vacía ===")
df_lsv["titulo_length"] = df_lsv["titulo"].str.len()
df_lsv["contenido_length"] = df_lsv["contenido"].str.len()
print(df_lsv[["titulo_length", "contenido_length"]].describe())

[4]

=== Text length statistics La Silla Vacía ===
          titulo_length  contenido_length
count      75.000000      75.000000
mean       64.960000     9934.946667
std        17.599478     5465.086864
min        14.000000      304.000000
25%        58.500000     6232.000000
50%        71.000000    10653.000000
75%        75.000000    12893.500000
max        94.000000    24664.000000
```

Fig. 6. Text length statistics La Silla Vacía

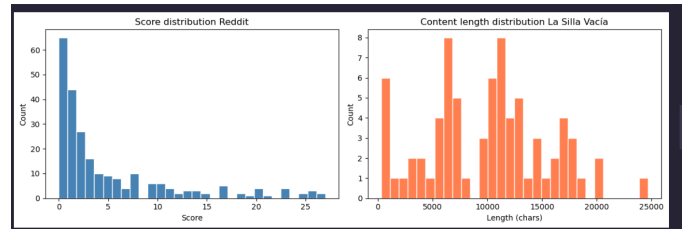


Fig. 7. Combined distribution analysis including Reddit score distribution and La Silla Vacía content length distribution.

The content length distribution of La Silla Vacía articles reveals substantial variability in article sizes, ranging from short publications to highly extensive journalistic pieces exceeding 24,000 characters. Figure 7 illustrates this variability and highlights the richness of the textual corpus used for storytelling analysis and keyword extraction tasks.

E. Reddit Score Distribution and Outlier Analysis

Reddit engagement scores were analyzed to identify anomalous or unusually popular posts.

```

print("=== Score outlier analysis Reddit ===")
Q1 = df_reddit["score"].quantile(0.25)
Q3 = df_reddit["score"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df_reddit[(df_reddit["score"] < lower) | (df_reddit["score"] > upper)]
print(f"Q1: {Q1}, Q3: {Q3}, IQR: {IQR}")
print(f"Lower bound: {lower}, Upper bound: {upper}")
print(f"Outliers: {len(outliers)} ({len(outliers)/len(df_reddit)*100:.2f}%)")

=== Score outlier analysis Reddit ===
Q1: 0.0, Q3: 7.0, IQR: 7.0
Lower bound: -10.5, Upper bound: 17.5
Outliers: 19 (7.82%)

```

Fig. 8. Score outlier analysis Reddit

The Reddit score distribution presents a strong right-skewed behavior, where most posts accumulate relatively low engagement scores while a smaller subset reaches significantly higher values. As illustrated in Figure 7, several posts exceed the interquartile upper bound, confirming the presence of highly engaging outlier discussions within the dataset. These anomalies may correspond to viral topics, controversial discussions, or highly relevant events during the observed collection period.

F. Temporal Coverage

The collected data spans approximately two months of activity across both sources.

```

print("=== Date range Reddit ===")
print(f"Min: {df_reddit['date'].min()}")
print(f"Max: {df_reddit['date'].max()}")

print("\n=== Date range La Silla Vacía ===")
print(f"Min: {df_lsv['fecha'].min()}")
print(f"Max: {df_lsv['fecha'].max()}")

=== Date range Reddit ===
Min: 2026-04-05 04:11:31
Max: 2026-05-27 00:15:21

=== Date range La Silla Vacía ===
Min: 2026-04-02 05:05:00
Max: 2026-05-26 18:35:27

```

Fig. 9. Date range Reddit and La silla vacia

The collected datasets provide consistent temporal coverage across both information sources, enabling the development of longitudinal analyses and storytelling-oriented aggregations over time. The Reddit dataset spans from April 5, 2026, to May 27, 2026, while the La Silla Vacía dataset covers the period from April 2, 2026, and May 26, 2026.

This temporal alignment is particularly valuable because it allows comparative analysis between social media discussions and digital journalism publications occurring during approximately the same time window. As a result, the pipeline supports the identification of content trends, engagement fluctuations, and variations in publication activity throughout the observed period.

V. GOLD LAYER ARCHITECTURE

The Gold layer represents the final analytical stage of the pipeline and is responsible for generating business-oriented

datasets optimized for governance monitoring, storytelling exploration, and trend analysis. Unlike the Bronze and Silver layers, which primarily focus on data ingestion and cleansing, the Gold layer transforms processed records into aggregated analytical outputs ready for reporting and decision-making purposes.

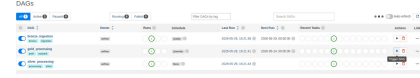


Fig. 10. DAGs run in Airflow

The Gold pipeline was implemented using Apache Airflow for orchestration and PySpark for scalable distributed processing. The workflow consumes all validated Silver-layer parquet files and applies multiple transformations and aggregations to generate consolidated analytical datasets.

The Gold layer produces two major categories of outputs:

governance-oriented datasets,
storytelling-oriented datasets.

These datasets are stored as parquet files inside the `data_lake_gold` directory to ensure efficient storage, interoperability, and compatibility with analytical tools.

A. Gold Processing Workflow

The Gold processing pipeline is composed of two main Airflow tasks:

`governance_summary()`
`storytelling_summary()`

Both tasks operate on Silver-layer parquet datasets and generate aggregated outputs independently.

The pipeline follows the following workflow:
Read all Silver parquet datasets.
Standardize schemas across sources.
Merge datasets when required.
Apply PySpark aggregations.
Generate governance metrics.
Generate storytelling datasets.
Export final parquet outputs to the Gold layer.

This modular design improves maintainability and enables future scalability for additional analytical tasks.

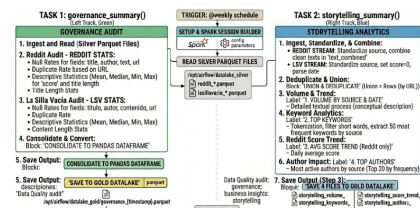


Fig. 11. Gold processing workflow

The Gold layer processes the validated Silver datasets to generate analytical outputs focused on governance monitoring and storytelling analysis. First, all parquet files from Reddit and La Silla Vacía are loaded using PySpark. The datasets are then standardized to ensure compatible schemas between both sources.

After standardization, the pipeline applies several aggregation and transformation operations such as grouping, filtering, deduplication, keyword extraction, and statistical computations. These operations generate consolidated analytical datasets that summarize publication activity, engagement behavior, author participation, and data quality indicators.

Finally, the generated datasets are exported as parquet files into the `datalake_gold` directory, allowing efficient storage and future analytical consumption.

The workflow was designed to maintain scalability, reproducibility, and compatibility with distributed data processing environments.

B. Governance KPI Architecture

The governance component of the Gold layer was designed to evaluate the quality, completeness, consistency, and reliability of the processed datasets. Governance indicators provide quantitative evidence about the health of the pipeline and the analytical usability of the data.

The governance task processes Reddit and La Silla Vacía datasets independently and computes multiple quality-related metrics using PySpark transformations.

The resulting governance dataset includes the following indicators:

These indicators enable continuous monitoring of data quality and provide traceability for future pipeline validation processes.

TABLE II
KEY PERFORMANCE INDICATORS (KPIs) AND DESCRIPTIONS.

KPI	Description
total_records	Total number of processed records. Measures the total volume of processed data and verifies that ingestion and transformation stages were completed successfully.
null_rate_title	Percentage of missing titles. Evaluates the completeness of title fields, which are essential for content identification and keyword analysis.
null_rate_author	Percentage of missing authors. Ensures contributor information is available for author activity analysis and traceability.
null_rate_text	Percentage of missing content. Validates the availability of meaningful textual content required for storytelling and NLP processing.
null_rate_url	Percentage of missing URLs. Confirms that records maintain traceability to their original data sources.
duplicate_rate	Percentage of duplicate records. Detects redundant records that could distort aggregations and analytical results.
score_mean	Average Reddit engagement score. Measures average Reddit engagement to evaluate overall community interaction levels.
score_median	Median Reddit engagement score. Provides a more robust central tendency metric less affected by outlier posts.
score_min	Minimum score value. Identifies the lowest engagement levels present in the dataset.
score_max	Maximum score value. Detects highly engaging or potentially viral content within Reddit discussions.
title_len_mean	Average text length. Evaluates the average size of textual content and helps characterize dataset richness.
title_len_min	Minimum text length. Identifies extremely short records that may contain limited analytical value.
title_len_max	Maximum text length. Detects large textual records that may represent detailed articles or extensive discussions.
computed_at	Timestamp of metric generation. Records the exact timestamp when governance metrics were generated, supporting reproducibility and auditing processes.



Fig. 12. Governance metrics output

C. Storytelling Analytics Architecture

The storytelling component of the Gold layer focuses on generating aggregated datasets capable of supporting exploratory analysis and narrative-oriented insights.

To achieve this objective, records from Reddit and La Silla Vacía were standardized and merged into a unified analytical structure using PySpark transformations. The storytelling pipeline generates four analytical datasets.

1) *Volume Trend Dataset*: The first aggregation analyzes publication activity over time.

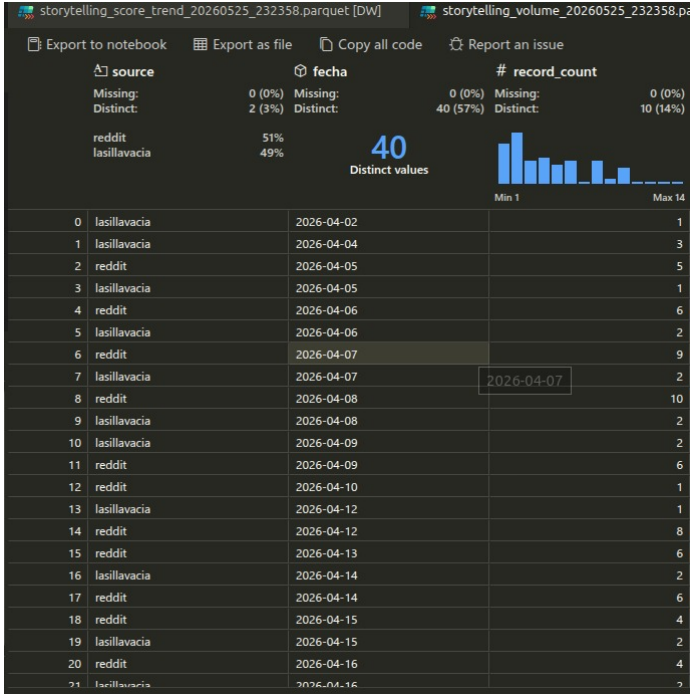


Fig. 13. storytelling_volume parquet

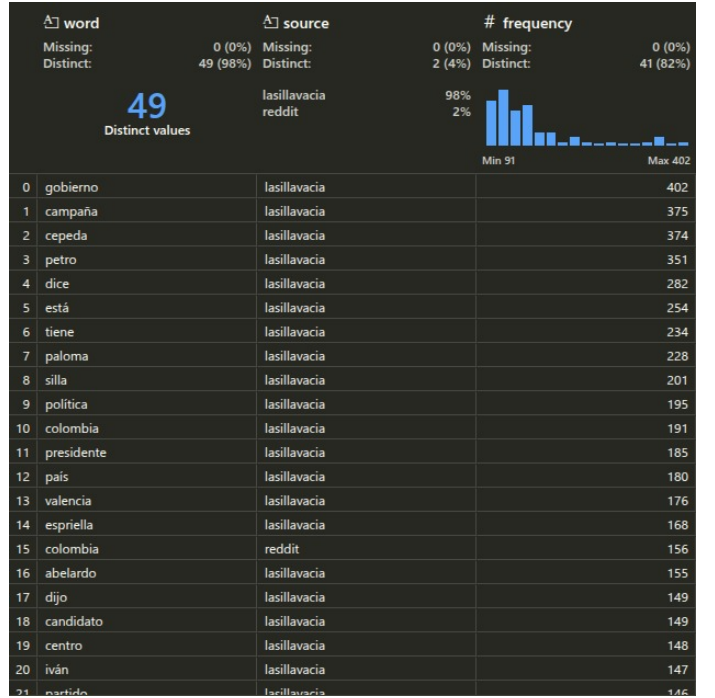


Fig. 14. storytelling_keywords parquet

This aggregation analyzes how publication activity changes over time across both sources. The dataset groups records by source and publication date to calculate the number of collected records per day. The resulting dataset enables temporal trend analysis and helps identify periods with higher publication activity or discussion intensity.

2) *Keyword Frequency Dataset*: The second aggregation extracts and counts the most frequent words from the cleaned textual content.

3) *Reddit Engagement Trend Dataset*: The third aggregation evaluates Reddit engagement behavior over time using average score analysis.

TABLE III
GENERATED FIELDS

Field	Description
source	Data source
fecha	Publication date
record_count	Number of records

This aggregation extracts the most frequent words from the cleaned textual content generated during Silver processing. PySpark tokenization and counting operations are applied to identify dominant terms across Reddit and La Silla Vacía. The resulting dataset supports thematic exploration and provides insights into the most discussed topics within the collected data.

TABLE IV
GENERATED FIELDS

Field	Description
fecha	Publication date
avg_score	Average Reddit score
post_count	Number of analyzed posts

This aggregation evaluates Reddit engagement behavior over time by calculating the average score and number of posts per date. The objective is to analyze how user interaction evolves throughout the observed collection period. The resulting dataset helps identify peaks of engagement and highly active discussion periods.

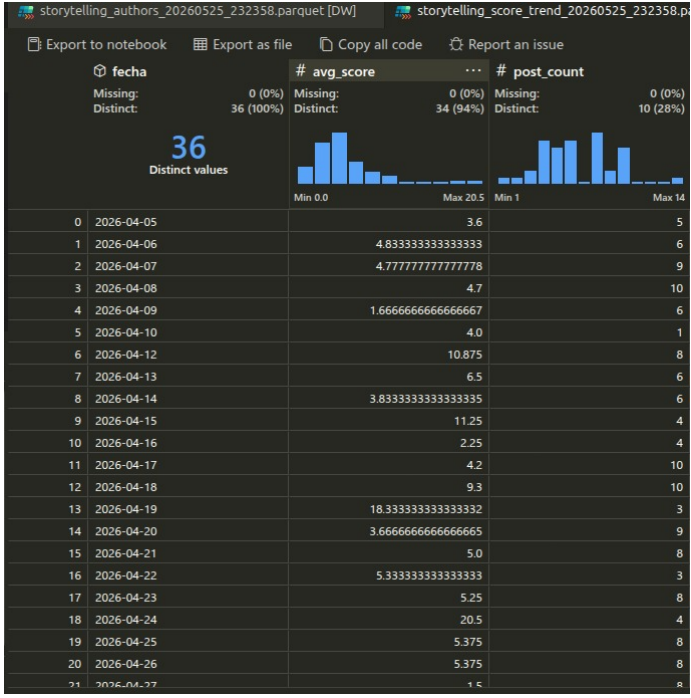


Fig. 15. storytelling_score_trend parquet

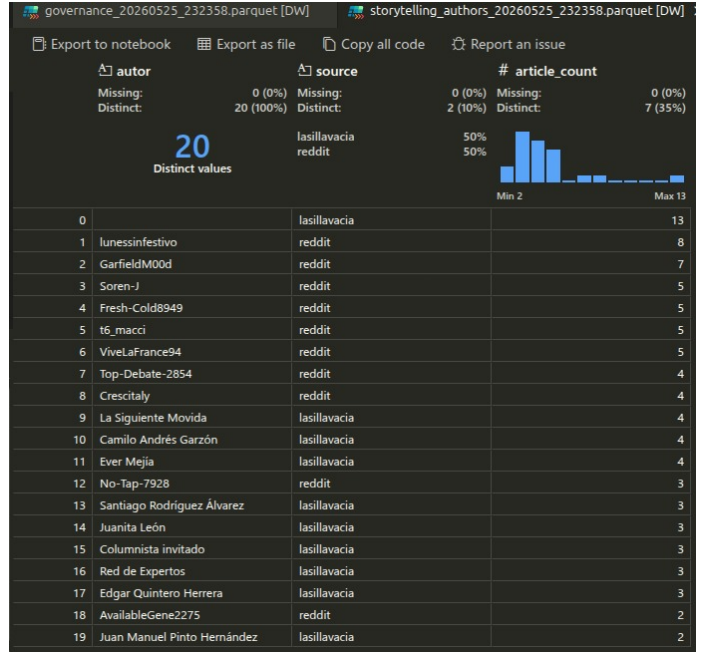


Fig. 16. storytelling_authors parquet

4) *Author Activity Dataset*: The final aggregation identifies the most active authors across both sources.

TABLE V
GENERATED FIELDS

Field	Description
autor	Author name
source	Data source
article _{count}	Number of publications

This aggregation identifies the most active authors from both sources by counting the number of publications associated with each contributor. The resulting dataset facilitates contributor analysis and highlights the users or journalists with the highest participation levels in the dataset.

5) *Storytelling Aggregations Justification Based on User Stories*: The storytelling aggregations were designed to address the analytical needs identified in the user stories defined during Workshop 1. Each aggregation supports a specific type of decision-making process or analytical exploration for the target users.

TABLE VI
STORYTELLING AGGREGATIONS JUSTIFICATION BASED ON USER STORIES

Aggregation	Related User Story	Justification
Volume Trend Dataset	Political Analysts, Journalists, Data Analysts	Identifies publication activity changes over time and detects periods with increased political discussion or media attention.
Keyword Frequency Dataset	Journalists, Social Researchers, Communication Strategists	Supports thematic exploration by identifying the most discussed topics and recurring narratives across both sources.
Reddit Engagement Trend Dataset	Political Analysts, Communication Strategists	Analyzes public interaction intensity through Reddit scores and detects highly impactful discussions.
Author Activity Dataset	Journalists, Social Researchers	Identifies the most active contributors and supports author participation analysis.
Governance KPI Dataset	Data Analysts, Policy Makers	Ensures data quality, consistency, and reliability before analytical or reporting usage.

The implemented storytelling architecture directly supports the information needs identified during the initial requirements analysis trend detection, sentiment and discourse explorationreporting support, analytical monitoring, contributor analysis and decision-support processes.

By connecting storytelling outputs with user-oriented analytical requirements, the Gold layer transforms processed data into actionable analytical information aligned with real-world decision contexts.

After viewing the dashboard, the functional user should be able to identify periods with increased discussion or publication activity, the most relevant and recurring topics, highly engaging Reddit conversations, the most active contributors and overall trends in political and social discourse.

VI. PYSPARK IMPLEMENTATION AND PROCESSING STRATEGY

PySpark was used as the main processing engine for the Gold layer because it provides efficient handling of large datasets, scalable transformations, and native compatibility with parquet-based Data Lake architectures. Its integration with Apache Airflow also allowed the pipeline to execute automated analytical workflows inside the Docker environment.

The Gold processing stage begins by creating a SparkSession, which acts as the entry point for all distributed processing operations. The configuration was optimized for local execution while still supporting parallel processing through multiple worker threads.

```
SparkSession.builder
.master("local[*]")
.appName("GoldStorytelling")
.config("spark.driver.memory", "1g") (
.config("spark.sql.shuffle.partitions", "4")
```

Additional parquet compatibility settings were also included to avoid timestamp serialization and datetime conversion issues between Pandas, PySpark, and parquet files.

Several PySpark operations were used during Gold-layer processing in the table VI.

TABLE VII
OVERVIEW OF PYSARK TRANSFORMATIONS USED

Transformation	Purpose
groupBy()	Dataset aggregation
agg()	Metric computation
dropDuplicates()	Duplicate elimination
union()	Dataset merging
explode()	Keyword extraction
split()	Text tokenization
mean()	Statistical aggregation
count()	Volume analysis

All Gold outputs were exported in parquet format to guarantee efficient storage, optimized analytical querying, compatibility with Spark ecosystems and reproducibility of analytical workflows. The use of parquet also reduced storage overhead while improving interoperability between different processing stages.

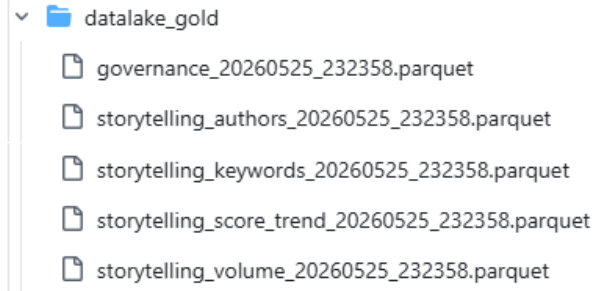


Fig. 17. Gold parquet folder structure.

VII. RESULTS AND ANALYTICAL INSIGHTS

Implementing the Gold layer successfully generated multiple analytical datasets that can support governance monitoring and exploration for storytelling purposes. The resulting outputs demonstrate how raw web data can be transformed into structured analytical assets via scalable processing pipelines.

A particularly notable outcome was the successful integration of diverse data sources into a unified analytical environment. Despite their structural differences, the implemented standardisation processes enabled consistent aggregations and comparative analysis of Reddit and La Silla Vacía data.

The governance dataset confirmed the effectiveness of the Silver-layer cleaning strategies. Null rate analysis showed complete data availability across critical attributes, while duplicate analysis reported a duplication rate of 0 percentage for both sources. These results demonstrate that the implemented validation mechanisms significantly improved the reliability of the dataset before Gold-layer processing.

From an analytical perspective, the storytelling datasets revealed meaningful insights into publication activity, engagement behaviour and thematic trends. Volume aggregation enabled temporal comparisons between Reddit discussions and digital journalism publications, and keyword extraction highlighted dominant themes in the collected corpus.

Additionally, Reddit score analysis identified high-engagement outliers, indicating that certain discussions generated substantially more community interaction than average. Similarly, author aggregation facilitated the identification of highly active contributors across both platforms.

VIII. CONCLUSIONS

This workshop successfully demonstrated the implementation of a complete analytical data pipeline based on a Medallion architecture comprising Bronze, Silver and Gold layers. By integrating Apache Airflow, PySpark, Pandas and Parquet-based storage, the project transformed heterogeneous web data into structured analytical datasets suitable for governance monitoring and storytelling exploration.

Improvements to the Silver layer significantly enhanced data quality by incorporating stricter validation mechanisms, enhanced deduplication strategies, timestamp normalisation and PySpark processing compatibility optimisations. These

refinements directly improved the reliability and consistency of Gold-layer outputs.

The Gold layer successfully generated governance KPIs and storytelling aggregations that can support exploratory analysis and analytical reporting. The resulting datasets provided insights into publication activity, engagement behaviour, thematic trends and author participation across Reddit and La Silla Vacía.

Alongside the technical implementation, the project emphasised the importance of iterative experimentation in modern data engineering workflows. The continuous refinement of pre-processing and validation strategies was essential for producing reliable analytical outputs from noisy and heterogeneous web data sources.

IX.

REFERENCES

- [1] Reddit, “Reddit API Documentation,” [Online]. Available: <https://www.reddit.com/dev/api/>. [Accessed: Mar. 2026].
- [2] Reddit Inc., “Reddit Content Policy,” [Online]. Available: <https://www.redditinc.com/policies/content-policy>. [Accessed: Mar. 2026].
- [3] K. Reitz, “Requests: HTTP for Humans,” Python Software Foundation, [Online]. Available: <https://docs.python-requests.org/>. [Accessed: Mar. 2026].
- [4] L. Richardson, “Beautiful Soup Documentation,” [Online]. Available: <https://www.crummy.com/software/BeautifulSoup/>. [Accessed: Mar. 2026].
- [5] Microsoft, “Playwright Documentation,” [Online]. Available: <https://playwright.dev/python/docs/intro>. [Accessed: Mar. 2026].
- [6] W. McKinney, “pandas: Powerful Python data analysis toolkit,” [Online]. Available: <https://pandas.pydata.org/docs/>. [Accessed: Mar. 2026].
- [7] R. Vink, “Polars Documentation,” [Online]. Available: <https://polars.github.io/polars/>. [Accessed: Mar. 2026].
- [8] D. Crockford, “The JSON Data Interchange Format,” RFC 8259, 2017.
- [9] R. Mitchell, *Web Scraping with Python: Collecting More Data from the Modern Web*, 2nd ed. O’Reilly Media, 2018.
- [10] T. White, *Hadoop: The Definitive Guide*, 4th ed. O’Reilly Media, 2015.
- [11] Apache Software Foundation, “Apache Airflow Documentation,” [Online]. Available: <https://airflow.apache.org/docs/>. [Accessed: Mar. 2026].